Bistatic UAV-Based 3D SAR Imaging Using Integrated Sensing and Communications Transceivers

Indraprastha Institute of Information Technology Delhi, New Delhi 110020 India

***Abstract*—**

***Index Terms*—multi-armed bandit, joint radar communication, upper confidence bound, analog beamforming**

I. Introduction

Unmanned Aerial Vehicless (UAVs) equipped with synthetic aperture radar (SAR) are becoming increasingly important platforms for applications related to remote sensing, surveillance, and situational awareness [1], [2]. UAV-based SAR systems combine platform agility, fast deployment, and lower cost with the key benefits of SAR, which include all-weather, day-night imaging and wide-area coverage. The ability of UAV to fly at low altitudes and hover over areas of interest allows them to collect high-quality imagery for disaster response, precision agriculture, and security [3], [4]. It is particularly useful in specific situations where traditional space-borne SAR is limited by long satellite revisit times and when manned aircraft are difficult, unsafe, or expensive to operate. Furthermore, recent advances in compact millimeter wave (mmWave) transceivers, lightweight antennas, onboard processing, and autonomous operation enable the integration of high-performance SAR payloads on small-to-medium UAVs. The large available bandwidth at mmWave frequencies provides fine-range resolution, enabling high-resolution imaging [5].

Monostatic UAV SAR systems, in which the transmitter and receiver are on the same platform, are the most popular SAR systems and are easiest to deploy in terms of synchronization. However, they are limited to facilitating two-dimensional imaging along range and cross-range. Interferometric, bistatic and multistatic configurations exploit additional spatial, Doppler, and scattering diversity enhancing radar detection metrics and can be configured for three-dimensional scene reconstruction. The theoretical foundations of bistatic image formation through inversion have been extensively studied in [6]. Some recent developments in 3D SAR imaging include interferometric SAR (InSAR) to map heights of structures [7], differential InSAR to monitor deformations [8] on the Earth's surface, and SAR tomography (TomoSAR) to resolve multiple scatterers along the elevation axis- which is also the focus of our work. Early demonstrations of TomoSAR used multiple airborne baselines to synthesize a second aperture across the tracks for optimally reconstructing elevation profiles [9]. Later developments used multipass spaceborne designs, capable of interferometry, focusing on long-term repeat-pass data to allow 3D focusing [10]. To specifically address temporal decorrelation, a set of differential TomoSAR schemes that rely on multidimensional spectral estimation and multifaceted signal classification algorithms has been introduced in [11]. More recent methods use holographic SAR tomography, which employs circular airborne trajectories and iterative adaptive reconstruction to provide super-resolution elevation reconstruction with restricted passes [12]. While these advances have significantly improved the 3D imaging capability of SAR systems, their practical realization on distributed UAV platforms introduces several implementation challenges. Among these, maintaining synchronization between spatially separated platforms is one of the most critical requirements for coherent SAR image formation.

Existing synchronization approaches for bistatic UAV SAR are broadly divided into two categories. The first class focuses on hardware solutions, including specialized microwave links, reference beacons, pilot exchanges, and precise timing sources based on global navigation satellite systems (GNSS) or atomic oscillators. Alternatively, to handle weight restrictions on UAV platforms, the second approach uses signal-processing-based algorithms that estimate and compensate for residual timing and phase errors directly from the received echoes. This approach has a lower hardware load but is effective only with high SNR and stable geometries [].

More recently, a third approach has been investigated, in which UAVs are augmented with communication links to address synchronization and data-handling challenges in a distributed architecture. In [13], UAV-to-ground and UAV-to-UAV connections are used to facilitate real-time data transfer and sensing-aware formation control for bistatic InSAR [13]. Coherent multistatic UAV SAR architectures have also been explored in [14], in which a direct line-of-sight sidelink is used as a phase reference for hardware synchronization. But these approaches do not scale to tomographic or 3D localization. Bistatic TomoSAR has more recently been achieved by repurposing an uplink communication signal to reconstruct tomographic elevation images [15]. The authors of [16] used orthogonal frequency-division multiplexing (OFDM) pilots over line-of-sight links to provide timing and phase references for bistatic SAR processing at a ground-based station. Specialized two-way synchronization chains with pulsed carrier exchanges have also enabled high-precision bistatic InSAR in-flight experiments [17]. All these studies have focused on the

coexistence of radar and communication systems, in which communication waveforms are used for synchronization, while separate conventional radar waveforms are employed for SAR imaging. It is worth noting that, instead of using auxiliary communication platforms as discussed above, it is advantageous to consider transceivers with integrated sensing and communication (ISAC) capabilities on a common hardware, waveform, and spectrum to manage synchronization and interference between the two.

Recent studies have explored waveform co-design strategies to enable joint wireless communication and high-resolution SAR imaging [18], [19]. While these approaches demonstrate the feasibility of dual-functional systems, they are predominantly realized in either monostatic configurations or rely on centralized or ground-based infrastructure. As a result, they do not address the challenges of distributed bistatic synchronization, particularly in UAV-based tomographic SAR scenarios.

Such approaches do not fully exploit the potential of unified ISAC operation, in which a single waveform, shared RF hardware, and common spectral resources are jointly used for both sensing and communication. In addition, most prior work is limited to two-dimensional imaging or simplified point-target models and does not address three-dimensional localization of complex, distributed targets within an ISAC framework. Furthermore, many studies assume stationary or ground-based receivers and simplified motion models, which limit their applicability to realistic UAV deployments involving mobile platforms, elevated terrain, and dynamic flight trajectories. In practical bistatic SAR systems, platform and target motion can introduce significant image displacement, defocusing, and localization errors whose severity depends on the bistatic geometry and target kinematics [20], [21]. In our proposed work, we overcome the above limitations by introducing a unified ISAC-based bistatic TomoSAR framework that leverages a single waveform, shared hardware, and common spectral resources for both communication-assisted synchronization and radar sensing, while enabling accurate 3D localization under realistic UAV operating conditions.

The key contributions of the proposed work, at the architecture, algorithmic, and system levels, are summarized as follows: (1) *Architecture Innovations:* A unified bistatic TomoSAR architecture is developed on a mmWave 802.11ad ISAC platform, with shared RF hardware, and common spectral resources for both communication and radar sensing. Here, communications assist in synchronization by exploiting the embedded information within the transmitted waveform, removing the need for dedicated synchronization links, auxiliary sensors, or additional RF hardware while enabling coherent bistatic operation. (2) *Algorithmic Innovations:* A novel geometric 3D localization approach based on sphere-plane intersection is proposed to accurately estimate scatterer positions in bistatic tomographic geometry, enabling reliable reconstruction of complex and distributed targets. The approach is benchmarked against the popular frequency modulated continuous wave radar signal using InSAR signal processing methods.

(3) *High Fidelity Simulation Framework:* To validate the proposed architecture and algorithm, we have used a high fidelity simulation model that incorporates realistic extended target on uneven terrains through digital elevation models, with shadowing and speckle. Further, the UAV trajectory model incorporates translational and rotational platform motion errors along with sensor errors. (4) *Scientific insights on interplay between sensing and communication:* The use of ISAC waveform instead of separate radar and communication systems result in a unique interdependence/interplay between the performance metrics of both functionalities. Specifically, we study how communication modulation schemes affect the radar localization metrics.

The proposed ISAC-Bistatic method consistently outperforms all other benchmark configurations (namely FMCW Bistatic, ISAC-InSAR, and conventional FMCW-InSAR) across all platform velocity noise levels, both with and without motion compensation. Specifically, under motion-compensated conditions and a high platform velocity standard deviation (8 m/s), the proposed method achieves a 84.3% reduction in target elevation (Z) RMSE (1.77 m) when compared to the conventional FMCW-InSAR baseline (11.22 m). Similarly, under a low velocity standard deviation (0.5 m/s), the proposed method maintains an elevation RMSE of 0.58 m, representing a 81.6% reduction compared to the baseline (3.12 m).

The paper is organized as follows. In Section II, we present a comprehensive overview of the proposed ISAC-based bistatic UAVs system model. This is followed by a discussion of the signal model and processing, along with the novel 3D localization algorithm, in Section III. In Section IV, we present the high-fidelity simulation setup for evaluating the performance of the proposed framework, followed by the timing analysis. Next, we present the ISAC performance metrics for different scenarios in Section V, followed by an in-depth discussion of the novel scientific findings and a discussion on future works in Section.VI.

Notation: Continuous and discrete time signals are represented by $(\cdot)$ and $[\cdot]$ respectively. Scalar variables, vectors, and matrices are represented by x, $\mathbf{x}$, and $\mathbf{X}$, respectively. The superscript $*$ corresponds to the frequency-domain signal.

II. Proposed System Model

The proposed architecture is illustrated in Fig. 1. A right-handed Cartesian coordinate system is considered to represent the system model, where the x, y, and z-axes denote the ground-range (cross-track) direction, the azimuth (along-track) direction, and the altitude, respectively. We consider two UAV platforms, UAV 1 and UAV 2, operating over a common ground swath area and exploiting a shared ISAC waveform to jointly enable synchronization through direct communication and bistatic radar sensing of ground based targets. UAV 1 consists of a transceiver, whereas UAV 2 only has a passive receiver. Both UAVs follow approximately linear trajectories parallel to the azimuth direction, configured in a stripmap SAR mode, where the radar antenna beam of

each UAV is fixed relative to the platform and continuously illuminates the ground swath during flight. UAV 1 and UAV 2

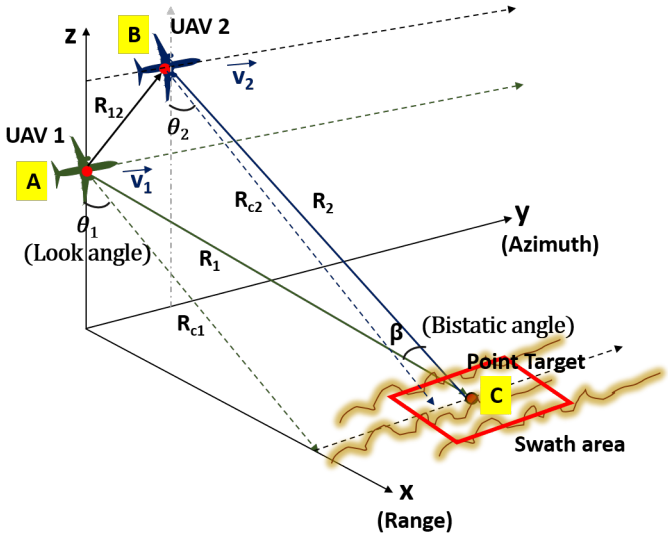


Fig. 1: System model of bistatic UAV

observe the scene with look angles, θ_1 and θ_2, respectively. The slant-range distances to the scene center from the two UAVs are R_{c1} and R_{c2} in Fig. 1. The bistatic angle at the target is β. The spatial separation between the two UAVs introduces angular, Doppler, and RCS diversity that is exploited for bistatic tomographic reconstruction.

The ISAC waveform chosen in this work is the IEEE 802.11ad physical layer (PHY) packet that was originally proposed for mmWave vehicular communications [22]. In prior work, a detailed description of the 802.11ad-based ground-based monostatic ISAC system architecture for automotive scenarios was presened [23], [24]. In this section, we discuss the adaptation of the ISAC framework for the bistatic aerial SAR scenario as shown in Fig. 2.

First, UAV 1 transmits packets: $\mathbf{x}_1, \mathbf{x}_2, \cdots \mathbf{x}_P$, in the mmWave band as shown in the top row labeled UAV 1-TX in Fig. 2. Each packet spans one pulse repetition interval T_{PRI} and the time span of P packets forms a coherent processing interval, T_{CPI}. Each UAV is equipped with two directional transmit antennas: one antenna is oriented toward the other UAV to facilitate communication-assisted synchronization, while the second antenna is directed toward the ground to illuminate the swath area for radar sensing. Thus, the single transmission signal serves simultaneously as a communication signal from UAV 1 to UAV 2 and as a radar-sensing waveform of ground-based targets. The backscattered radar echoes from the ground swath are collected by the receiver on UAV 1 (monostatic channel), represented as $\mathbf{y}_p^1$. This operation is represented by the second row labeled UAV-RX in Fig. 2. The direct-path communication signal $\mathbf{y}_p^{dir}$ and the bistatic reflected signal $\mathbf{y}_p^{ref}$ are acquired by UAV 2 as illustrated in the UAV 2-RX row in Fig. 2. As shown in the UAV 2-Comm row in Fig. 2, the $\mathbf{y}_p^{dir}$ signal is processed to synchronize the two UAVs. Further, each UAV performs local 2D SAR processing using its own received echoes as shown in the UAV 1 and UAV 2 RSP rows in Fig. 2.

A. Communication-Based Synchronization of Bistatic Nodes

In the proposed ISAC-enabled bistatic UAV framework, the communication-assisted synchronization mechanism presented in [25] should we omit? is adopted to achieve phase-coherent bistatic processing without requiring separate dedicated synchronization hardware or external timing references. We assume that the UAV nodes know the idealized trajectories aprior and the objective is to correct for timing, phase and frequency offsets introduced by trajectory errors and hardware impairments. Each packet of IEEE 802.11ad comprises a preamble, header, a variable-length data payload, and an optional beamforming training field appended for antenna weight vector optimization which is omitted here since the ISAC transceiver is operated in the stripmap SAR configuration with a fixed look angle. The preamble operates at a baseband chip rate of 1.76 Gsps and is partitioned into a Short Training Field (STF) and a Channel Estimation Field (CEF). Both are composed of short and long duration repeated Golay sequences, respectively, chosen for their ideal impulse-like auto-correlation and zero periodic cross-correlation properties. UAV 2 estimates the packet start and coarse time offset by evaluating the auto-correlation metric of the received STF from $\mathbf{y}_p^{dir}$. Simultaneously, the phase of the correlation peak directly reveals the coarse frequency offset. Then the node performs a mismatched cross-correlation with local replicas of the complementary pairs within the CEF of $\mathbf{y}_p^{dir}$. Here, the sidelobes of the resulting channel impulse response are analytically canceled out and the localized, zero-sidelobe peak indicates the timing offset between the two independent UAV nodes. Subsequently, these offsets are digitally compensated yielding a fully coherent bistatic radar data cube for joint target localization and tracking.

B. Bistatic SAR and 3D Localization

Specific Golay sequences within the CEF also serve as the radar waveform for ISAC and the ordering of these sequences across multiple packets are chosen to ensure their suitability for identifying moving targets as described in [26], [27]. UAV 1 forms a conventional two-dimensional monostatic SAR image in range and crossrange by processing $\mathbf{y}_p^1$. UAV 2 processes the echoes, $\mathbf{y}_p^2$, using the known bistatic geometry and the synchronization parameters obtained from the communication packet. At both platforms, motion compensation is applied to correct for translational and rotational deviations in the actual UAV trajectories, ensuring phase coherence under realistic flight dynamics.

The range and cross-range information estimated by UAV 1 from the obtained 2D SAR image over the P packets comprising T_{CPI} is embedded in the data field of a subsequent $(P+1)^{th}$ packet (highlighted by red data block in the first row in Fig. 2) and transmitted to UAV 2. This strategy avoids raw-data transfer while enabling joint bistatic processing. The combined monostatic and bistatic observations provide sufficient spatial diversity to enable 3D localization

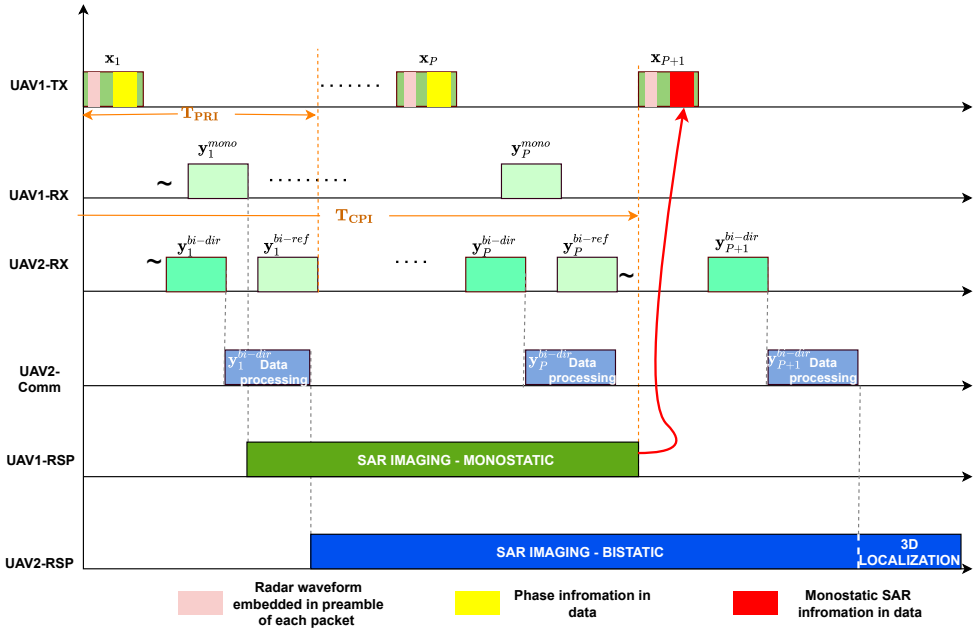


Fig. 2: System architecture of bistatic UAV framework

of scatterers. Further, our proposed novel sphere-plane intersection algorithm is used to estimate scatterer height and position, enabling robust 3D reconstruction of complex targets and terrain at elevated altitudes.

III. Signal Model and 3D SAR Processing

This section presents the detailed signal model and SAR processing along with motion compensation framework for the proposed ISAC-based bistatic UAV system. In this work, we consider IEEE 802.11ad PHY packet transmission from UAV 1 to UAV 2 to enable communication-assisted synchronization, followed by 3D target localization over elevated terrain.

A. Signal Model

We first describe the signal generation process at the transmitter of UAV 1. Let $x[m]$ denote the discrete-time baseband representation of the radar component of the transmitted IEEE 802.11ad PHY. Each pth packet is transmitted once per T_{PRI}, and comprises M samples corresponding to the sampling interval T_s, expressed as

$$\mathbf{x}_p(t) = \sqrt{E_s} \sum_{m=0}^{M} \mathbf{x}[mT_s]\delta\left(t - mT_s - pT_{PRI}\right) e^{+j2\pi f_c t}. \quad (1)$$

Here $\delta(\cdot)$ is the Dirac-delta function in the time domain (t). The time symbols are scaled such that each symbol has energy E_s, and subsequently upconverted to the mmWave carrier frequency f_c. The upconverted analog waveform is then radiated through high-gain transmit antennas. The radar echoes are collected at UAV 1 after two-way propagation to the ground swath. After downconversion and digitization, the received baseband signal corresponding to the p-th PRI is

$$\mathbf{y}_p^{mono}[m] = \sum_{b=1}^{B} \alpha_1(b)\, \mathbf{x}_p\left[m - \frac{2R_{1_p}(b)}{c}\right] \exp\left(-j\frac{4\pi}{\lambda} R_{1_p}(b)\right) + \beta_1[m] + \eta_1[m], \quad (2)$$

where $\alpha_1(b)$ is the strength of the target returns from each bth target that includes the target's RCS and two-way path loss factor in the monostatic channel. $R_1(b)$ denotes the instantaneous slant-range distance between UAV 1 and the bth scatterer at slow-time index p. Further, c and λ are the speed of light and the carrier wavelength, respectively; and β_1 and η_1 denote ground clutter present in the environment and the complex additive white Gaussian noise at the receiver of UAV 1, respectively.

The signals transmitted by UAV 1 are received at UAV 2 through two distinct propagation mechanisms through separate antennas. The first component corresponds to the direct line-of-sight (LoS) one-way path from UAV 1 to UAV 2 from the first antenna mounted on UAV 2 oriented towards UAV 1. After downconversion and digitization, the received signal at UAV 2 from this direct path is are we consistent with the naming scheme with respect to previous section?

$$\mathbf{y}_p^{dir}[m] = \zeta\, \mathbf{x}_p\left[m - \frac{R_{12_p}}{c}\right] \exp\left(-j\frac{2\pi}{\lambda} R_{12_p}\right) + \eta_2[m], \quad (3)$$

where ζ and R_{12_p} are the one-way path loss factor and instantaneous distance between the two UAVs, respectively; and η_2 models the complex additive white Gaussian noise (AWGN) at the receiver of UAV 2. The second component of the received signal arises from two-way propagation of the transmitted signal to the ground, and reflected towards UAV 2 by the B ground scatterers. This is gathered by the second antenna mounted on UAV 2 which is oriented towards the

ground at look angle denoted in Fig.**??**. The bistatic received signal is expressed as

$$\mathbf{y}_p^{ref}[m] = \sum_{b=1}^{B} \alpha_2(b)\, \mathbf{x}_p \left[m - \frac{R_{1_p}(b) + R_{2_p}(b)}{c} \right] \exp\left(-j \frac{2\pi}{\lambda} \left(R_{1_p}(b) + R_{2_p}(b) \right) \right) + \beta_2[m] + \eta_2[m], \quad (4)$$

where $\alpha_2(b)$ denotes the complex scattering coefficient of the bth target, incorporating both the target radar cross-section (RCS) and the bistatic path loss factor; $R_{2_p}(b)$ is the instantaneous slant-range distance between UAV 2 and the bth target while β_2 accounts for ground clutter contributions. Subsequently, both signals are forwarded to the radar signal processing block, where they are jointly used for 3D localization, as described in the later section.

B. 3D SAR Processing

The radar echoes over each CPI are collected independently at the receivers of UAV 1 and UAV 2 and organized into two 2D matrices comprising fast-time and slow-time samples and processed to obtain corresponding monostatic and bistatic SAR images. In this work, the range-Doppler algorithm (RDA) is applied on the radar data, over other alternatives, due to its computational efficiency and suitability for stripmap UAV geometries [28]. The processing consists of motion compensation, range-compression, range cell migration correction (RCMC), and azimuth-domain focusing, as illustrated in Fig. 3.

Following data acquisition, a coarse MOCOMP stage is first applied to correct for translational and rotational deviations in the UAV trajectories, thereby preserving phase coherence across pulses. The coarse-compensated received baseband signal in the monostatic and bistatic channels can be expressed as,

$$\mathbf{s}_p^{mono}[m] = \mathbf{y}_p^{mono}[m] \exp\left(j \frac{4\pi}{\lambda} \Delta R_p^{mono} \right), \quad \mathbf{s}_p^{ref}[m] = \mathbf{y}_p^{ref}[m] \exp\left(j \frac{4\pi}{\lambda} \Delta R_p^{bi-ref} \right). \quad (5)$$

Here, ΔR_p^{mono} and ΔR_p^{bi-ref} denote the range compensation term for monostatic and bistatic channels respectively, which are estimated using onboard INS/GPS measurements, as described in detail in a subsequent section. Following this, range compression is performed via matched filtering in the range-frequency domain for monostatic and bistatic channels as

$$\mathbf{S}^*{}_{rc_p}{}^{mono}[f_r] = \mathbf{S}^*{}_p{}^{mono}[f_r]\, \mathbf{X}^*[f_r], \quad \mathbf{S}^*{}_{rc_p}{}^{bi-ref}[f_r] = \mathbf{S}^*{}_p{}^{bi-ref}[f_r]\, \mathbf{Y}^*{}_p{}^{mono}[f_r], \quad (6)$$

where $\mathbf{X}^*$ represents the spectrum of the transmitted waveform in the monostatic channel. An inverse transform yields the range-compressed signal $\mathbf{s}_{rc_p}{}^{mono}[r]$. In the bistatic channel, the received signal in the range-frequency domain is matched-filtered with the range-frequency converted direct signal $\mathbf{Y}^*{}_p{}^{mono}[f_r]$ to account for bistatic propagation effects. Further, range offset is being compensated by estimating the slant-range deviation using the position coordinates of UAV 1 obtained from the synchronization data embedded in the communication signal, thereby enabling accurate alignment of the range bins with respect to the reference geometry. Then, inverse transform yields the range-compressed and offset corrected signal as $\mathbf{s}_{rc_p}{}^{bi-ref}[r]$. During the capturing of the raw data from both UAVs, coupling between range and azimuth arises from the non-linear slant-range history of each scatterer due to the platform motion. This effect causes the energy of a target to migrate across multiple range bins over slow time, leading to defocusing if not corrected. RCMC aligns the target responses to a common reference range corresponding to the scene center, thereby ensuring that the energy of each scatterer is confined to a single range cell for both the monostatic and bistatic images prior to azimuth processing [29]. The resulting RCMC-corrected signal is can be expressed as $\mathbf{s}_{rc_p}^{\text{mono/bi-ref}}[r] \xrightarrow{\text{RCMC}} \mathbf{s}_{rcmc_p}^{\text{mono/bi-ref}}[r]$.

Azimuth processing is then performed by exploiting the Doppler modulation induced by platform motion. The RCMC-compensated data are transformed along the slow-time as $\mathbf{s}_{rcmc_r}^{\text{mono/bi-ref}}[p] \xrightarrow{\text{Azimuth Fourier transform}} \mathbf{S}^{\dagger}{}_{rcmc_r}^{\text{mono/bi-ref}}[f_{D_p}]$ and the azimuth compression is achieved using a Doppler-domain matched filter,

$$\mathbf{S}_{ac_r}^{\text{mono/bi-ref}}[f_{D_p}] = \mathbf{S}^{\dagger}{}_{rcmc_r}^{\text{mono/bi-ref}}[f_{D_p}]\, H^{\dagger}[f_D], \quad (7)$$

where $H^{\dagger}[f_{D_p}]$ corresponds to the azimuth reference signal that is designed according to the corresponding monostatic and bistatic geometries CHECK. Following the inverse Doppler transformation, a focused two-dimensional SAR image is obtained for each channel.

C. Proposed 3D Localization

From a geometric perspective, we consider two spheres in 3D, each centered at the respective UAV positions $\mathbf{A} = [x_A, y_A, z_A]$ and $\mathbf{B} = [x_B, y_B, z_B]$, corresponding to UAV 1 and UAV 2, respectively, while the unknown target position is given by $\mathbf{O} = [x_O, y_O, z_O]$. as illustrated in Fig. 4. The bistatic angle β corresponds to the angle subtended at the target between the two UAVs. These are time varying because UAVs are moving. Are we doing this every fast time sample, or every slow time sample or every CPI? Here y_O corresponding to the cross-range of the target is estimated from the 2D SAR maps of monostatic and bistatic channels. The radii of these spheres are R_1 and R_2, respectively and are each estimated from the range of the target from the monostatic and bistatic SAR images. The distance between the UAVs is R_{12}. The intersection of the two spheres forms a circle, and the plane obtained by subtracting the two sphere equations uniquely contains this intersection circle. Is this scheme consistent with the cartesian coordinates used in the previous section Note that the proposed formulation naturally integrates with the bistatic tomographic geometry and does not require multiple baselines or additional UAV platforms, thereby reducing system complexity while maintaining high localization accuracy.

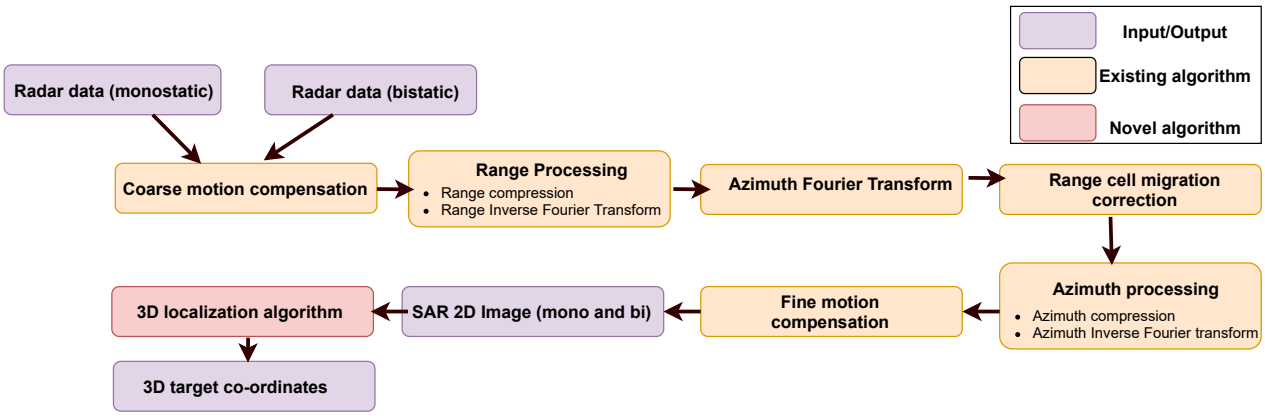


Fig. 3: Flowchart for 3D localization of target on an elevated ground

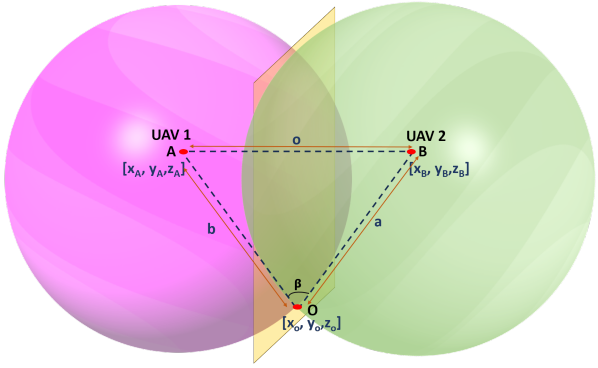


Fig. 4: 3D localization based on sphere-plane intersection algorithm

what are a and b? why are new variables being introduced? Using the cosine law, β can be obtained from

$$\cos\beta = \frac{[\mathbf{A}-\mathbf{O}]\cdot[\mathbf{B}-\mathbf{O}]}{R_1 R_2}. \tag{8}$$

Expanding the dot product yields

$$\begin{aligned}\cos\beta =&[x_A - x_O][x_B - x_O] + [y_A - y_O][y_B - y_O] \\ &+ [z_A - z_O][z_B - z_O] = ab\cos\beta.\end{aligned} \tag{9}$$

equation does not look right The SAR-derived slant ranges further impose spherical constraints on the target position. Accordingly, the target must satisfy

$$[x_O - x_A]^2 + [y_O - y_A]^2 + [z_O - z_A]^2 = b^2, \tag{10}$$

$$[x_O - x_B]^2 + [y_O - y_B]^2 + [z_O - z_B]^2 = a^2. \tag{11}$$

Subtracting (10) from (11) eliminates the quadratic terms in $[x_O, y_O, z_O]$ and results in a linear plane constraint,

$$Q_x x_O + Q_y y_O + Q_z z_O = D, \tag{12}$$

where

$$Q_x = 2[x_B - x_A], \tag{13}$$

$$Q_y = 2[y_B - y_A], \tag{14}$$

$$Q_z = 2[z_B - z_A], \tag{15}$$

and

$$D = b^2 - a^2 + x_B^2 - x_A^2 + y_B^2 - y_A^2 + z_B^2 - z_A^2. \tag{16}$$

The 3D target location $\mathbf{O}$ is then obtained by jointly solving the plane constraint in (12) and the angular constraint in (9). How? Since one of the resulting equations is quadratic, up to two candidate solutions may arise. The physically valid solution is selected based on platform–target visibility constraints and scene geometry. Lot of repetitive text in this section. Please rewrite for greater clarity.

D. Motion Errors and Motion Compensation Techniques

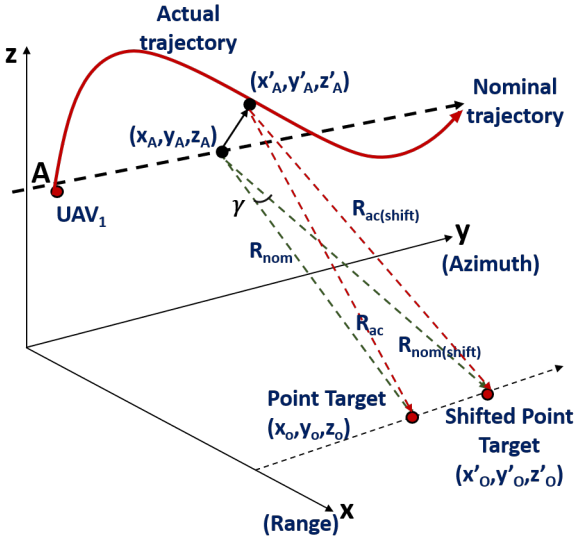


Fig. 5: Motion errors: translational and rotational

It is unclear where this is done... earlier only coarse mocomp is mentioned.

1) Motion Errors: Let $[x_A, y_A, z_A]$ denote the nominal UAV 1 position and $[x'_A, y'_A, z'_A]$ the actual UAV position in the 3D Cartesian plane. These positions vary with the slow time index as they continue to move with a constant velocity along the azimuth dimension. For a stationary point target

located at $[x_O, y_O, z_O]$, the nominal slant range for each pth slow time index can be written as

$$R_{\text{nom}}[p] = \sqrt{(x_A[p] - x_O)^2 + (y_A[p] - y_O)^2 + (z_A[p] - z_O)^2}. \tag{17}$$

Due to translational motion errors, the actual slant range deviates from the true slant range as shown:

$$R_{\text{ac}}[p] = \sqrt{(x'_A[p] - x_O)^2 + (y'_A[p] - y_O)^2 + (z'_A[p] - z_O)^2}. \tag{18}$$

This translational displacement introduces range-dependent phase errors in the received signal, resulting in defocusing along the range dimension.

Besides translational perturbation, rotational motion, particularly yaw and pitch, causes angular misalignment of the antenna beam with respect to the swath. As shown in Fig. 5, a rotational deviation by an angle γ shifts the apparent target position from $[x_O, y_O, z_O]$ to $[x'_O, y'_O, z'_O]$ which produces a cross-range displacement. Thus, the nominal cross-range of the target is effectively modified from R_{nom} to $R_{\text{nom}}^{\text{shift}}$, introducing azimuth-dependent phase errors that blur the image in the cross-range dimension. In the presence of both translational and rotational motion errors, distortions arise in both range and azimuth focusing, leading to inaccuracies in 3D localization. The combined effect of these errors can be represented through a shifted slant-range expression, as shown here,

$$R_{\text{ac}}^{\text{shift}}[p] = \sqrt{R_{\text{ac}}^2[p] + (x_O - x'_O[p])^2}, \tag{19}$$

where the cross-range displacement $x_O - x'_O[p]$ is related to the bistatic look geometry as

$$x_O - x'_O[p] = \tan\gamma[p] R_{\text{nom}}[p]. \tag{20}$$

These coupled translational and rotational effects introduce 2D geometric distortions and phase perturbations that severely impact SAR image quality and subsequent 3D localization. Therefore, accurate modeling and compensation of both motion components are critical for preserving phase coherence in bistatic UAV SAR systems. In this work, these errors are mitigated through a dedicated motion compensation (MOCOMP) framework, ensuring robust image focusing and reliable 3D target reconstruction under realistic UAV flight dynamics.

2) Motion Compensation Techniques: Next, we discuss the compensation techniques to autofocus the blurred image caused by motion errors. Phase coherence in UAV-based SAR systems requires accurate MOCOMP, especially in bistatic systems where the motion errors can catastrophically reduce the quality of image focus and 3D reconstruction. MOCOMP is applied in two consecutive phases in the proposed framework: *coarse motion compensation* followed by *fine motion compensation*. This two-step approach allows powerful correction of large-scale trajectory errors and residual phase errors that remain after the initial compensation.

Coarse Motion Compensation: Coarse MOCOMP is applied to the raw radar data before being sent to the SAR processing block. It aims to compensate for dominant phase errors caused by translational and rotational deviations of the UAVs from their nominal paths. The actual position of the UAV are obtained from GPS/INS measurements. Thus, in the case of a stationary target, the resulting range deviation for each slow time index can be written as

$$\Delta R(p) = R_{\text{ac}}^{shift}[p] - R_{\text{nom}}[p]. \tag{21}$$

This deviation introduces a phase error in the received signal, which is compensated by applying the coarse MOCOMP phase correction $\frac{4\pi}{\lambda}\Delta R(p)$ as shown in Eq. 5 for both monostatic and bistatic channels respectively. For the bistatic operation, $\Delta R[p]$ is computed as the sum of the transmitter-target and target-receiver range deviations, and the corresponding phase correction is applied accordingly. This coarse correction aligns the raw data to a common reference trajectory.

Fine Motion Compensation via Phase Gradient Autofocus: Despite coarse compensation using navigation data, residual phase errors remain due to inaccuracies in the GPS/INS measurements and unmodeled platform dynamics. These residual errors lead to image defocusing, degraded target localization accuracy, and loss of coherent integration gain. Therefore, a fine MOCOMP stage is applied after preliminary SAR image formation to estimate and compensate for the remaining phase distortions. Several autofocus techniques have been proposed in the SAR literature, including minimum-entropy autofocus [30], cross correlation autofocus [31], and phase gradient autofocus (PGA) [32]. More recently, sparse reconstruction and joint autofocus approaches have been investigated to simultaneously recover the target scene and motion-induced phase errors, particularly under undersampled measurements and severe motion uncertainties [33]. However, these methods generally involve iterative optimization procedures and significantly higher computational complexity. Among the available autofocus techniques, PGA has emerged as one of the most widely adopted approaches due to its computational efficiency, robustness, and ability to accurately estimate phase errors directly from the SAR image data. Furthermore, PGA does not require prior knowledge of the scene reflectivity and can effectively compensate for slowly varying residual phase errors commonly encountered in UAV-based SAR systems. Thus, in this work, fine MOCOMP is implemented using the phase gradient autofocus (PGA) algorithm [32].

Let $\mathbf{s_{ac_r}}[p]$ denote the coarse compensated, range and azimuth compressed for monostatic and bistatic channels. The residual phase error to be compensated is denoted by $\psi_{\text{res}}[p]$. The PGA algorithm estimates the phase gradient $\frac{d\psi_{\text{res}}[p]}{dp}$ by exploiting the phase consistency of dominant scatterers across azimuth. Integration of the estimated gradient yields the residual phase error $\psi_{\text{res}}[p]$, which is removed via the fine MOCOMP correction. This correction is applied iteratively until image sharpness converges. The combined application of coarse GPS-based MOCOMP and fine PGA-based autofocus provides a robust and practical motion compensation framework. Coarse MOCOMP ensures geometric alignment using external navigation data, while fine MOCOMP compensates residual, data-driven phase errors without requiring additional hardware or calibration targets. This two-stage strategy is applied consistently to both

monostatic and bistatic channels, preserving inter-channel coherence and enabling high-quality SAR focusing and reliable 3D localization under realistic UAV flight conditions.

IV. Simulation Setup

In this section, we describe the simulation environment where the proposed bistatic UAV framework based on ISAC is evaluated. The simulation models the radar system, UAV motion, extended ground-based targets, and uneven terrains with speckle using the digital elevation model (DEM).

A. Bistatic UAV System Parameters

Table I summarizes the radar system parameters used in the simulations. The design is aimed at obtaining high-resolution images without violating lightweight UAV payload requirements. Operating at mmWave frequency allows compact antenna design and finer range resolution, which is crucial in high-fidelity 3D imaging. The system has a range resolution of 0.08 m and a cross-range resolution of 0.3 m with a ground swath area of 100 m $\times$ 100 m. The pulse repetition interval is 2 μs, which corresponds to the duration of a single IEEE 802.11ad packet. UAVs 1 and UAVs 2 are positioned at slant ranges of 1 km and 2 km with respect to the swath center, respectively, from the center of the swath, under which flat-Earth assumptions hold. Both UAVs move with a platform velocity of 50 m/s along linear trajectories parallel to the azimuth direction. The look angles of UAVs 1 and UAVs 2 are 20° and 45°, respectively, to provide sufficient angular diversity for bistatic tomographic reconstruction. The squint angles are assumed to be 0° for both platforms. To balance the stripmap geometry and reduce the computational complexity of the Doppler centroid. what does this mean? What about the motion errors

TABLE I: Bistatic UAV SAR System Parameters

Parameter	Value
Center frequency	35 GHz
Range resolution	0.6 m
Cross-range resolution	0.3 m
Swath area	100 m $\times$ 100 m
Pulse repetition frequency (PRF)	1500 Hz
Pulse width	2 μs
Slant range to swath center (UAV 1, UAV 2)	1 km, 2 km
Platform velocity (UAV 1 and UAV 2)	50 m/s
Look angle (UAV 1, UAV 2)	20°, 45°
Squint angle (UAV 1, UAV 2)	0°, 0°

B. Target and Terrain Model

The target considered in this work is a realistic 3D automotive mid-size car model as seen in Fig. 6. The chassis, and wheels and other components of the vehicle are modelled with a metallic triangular mesh as described in [34], [35]. We consider that the centroid of each triangular facet corresponds to an individual scattering element, with radar cross-section determined by the facet's area, and aspect with respect to the radar. The total radar cross-section of the target is thus the superposition of the returns from all the facets and varies with time with a mean of 10 m^2. The target is modeled to move on aground swath at a location [25,25,10] m in 3D space is the car movign?. A realistic uneven ground

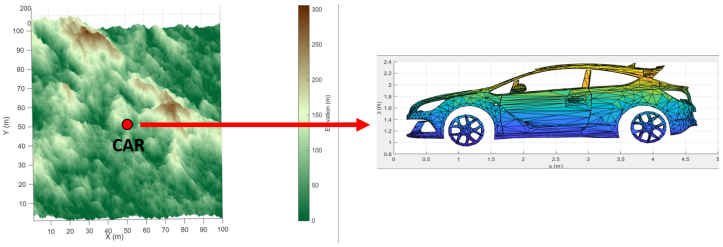


Fig. 6: Automotive target (mid-size car) mounted on a digital elevation terrain [FIGURE AXES TO BE FORMATTED]

terrain, varying from 0 to 20 m above sea level and spanning the swath, is modeled using the DEM provide details or can even say Mathworks tool box or something. The terrain occupies the swath area, with elevations varying from 0 m to 20 m to simulate non-flat topography. The DEM incorporates uneven surface features with varying RCS to induce realistic slant-range variations with geometric distortions - what does this mean? in the received echoes. The car is placed in high terrain to determine how the proposed sphere-plane intersection localization algorithm is able to recover the correct three-dimensional coordinates in non-planar conditions. In this subsection, we describe the methodology used to evaluate the performance of the proposed ISAC-based bistatic SAR framework and validate the 3D localization algorithm against conventional Inferometric SAR (InSAR) as a baseline []. For the bistatic 3D localization approach, monostatic and bistatic SAR image are used, obtained from two UAVs as described in the previous section. In contrast, for the InSAR approach, two monostatic SAR images are acquired using two UAVs with a vertical baseline separation of 10 m, while all other parameters remain the same. Is the baseline separation same for the bistatic scenario?

The experiments were conducted in MATLAB 2025a. To better reflect realistic conditions in our evaluation, we used Monte Carlo framework and introduced randomized perturbations to simulations. were the pertubations introduced in the error or noise or both To quantitatively evaluate the proposed 3D localization approach, we perform 1000 independent Monte Carlo experiments to validate our algorithm under varying platform and target conditions. In each experiment, various platform and target parameters are randomly varied to study their effect on our proposed method. Needs more details like what is the distribution (gaussian or uniform etc)? What are the parameters of the distribution (means, variance, etc?

First, the spatial coordinates of the automotive target (x, y, z) are uniformly distributed within the simulated 3D region of interest. It is an extended target and not a point target. Are you referring to the centroid of the car? What is the make and model? Note that the target can only be present at arbitrary positions within the illuminated swath and over elevated terrain. Second, the RCS of the target is modeled as a complex quantity whose magnitude has a mean value of

$10\ \mathrm{m}^2$, while the phase is uniformly distributed between 0 and 2π to represent random scattering behavior. Not clear. In the previous section, there was only a single RCS value for the extended target. Third, the look angles of the UAVs platforms are varied randomly between $0°$ and $45°$, which corresponds to typical ground observation geometries in UAVs-SAR systems. Again, in the previous section, all of these parameters were fixed Finally, the slant range between the UAVs platforms and the scene center is randomly selected within the interval of 800 m to 1200 m in order to account for variations in imaging geometry. Discrepencies between this section and simulation set up table.

For each experiment, the 3D coordinates of the target are estimated using the proposed bistatic processing framework. The localization accuracy is quantified using the root mean square error (RMSE) of the estimated coordinates relative to the ground-truth target position. The RMSE values are computed separately for the range (x), cross-range (y), and elevation (z) components corresponding to the highest peak in the reconstructed image. Further, the proposed ISAC-based waveform is compared with a conventional FMCW waveform to examine the impact of the choice of waveform on localization accuracy. Is the other waveform of similar carrier frequency and bandwidth? **The proposed 3D localization approach is also benchmarked against a conventional interferometric SAR (InSAR) method in order to highlight the advantages of the bistatic tomographic geometry. HAS THIS ALREADY BEEN SAID?**

To investigate the robustness of the localization algorithm to measurement uncertainties, the RMSE performance is evaluated for increasing errors in platform translational velocity, modeled as Gaussian perturbations with standard deviation swept over $[0.25, 0.5, 1, 2, 4, 8]$ in a log-spaced manner.

V. Performance Analysis

A. Timing Analysis of the Proposed Architecture

The timing of the proposed ISAC-based bistatic UAV framework depicted in Fig. 2 is analysed to measure the end-to-end latency of communication-assisted synchronization, SAR image formation, and 3D localization. The total latency of the framework is based on the number of packets, their duration as well as the processing time. In this work, all the processing is carried out in floating-point in MATLAB 2025a on an Intel Core i7 processor with 128 GB RAM.

It is worth noting that several conventional synchronization approaches for the bistatic system reported in the literature, including GNSS-disciplined oscillators, chip-scale atomic clocks, two-way synchronization links, and direct-path sidelink synchronization [14], [17], [36], [37], can achieve synchronization accuracies ranging from tens of nanoseconds down to sub-nanosecond levels. However, such approaches typically require additional synchronization hardware, dedicated RF chains, external reference oscillators, or auxiliary communication links, thereby increasing the payload complexity, power consumption, and overall system cost of lightweight UAV platforms. THIS PARAGRAPH READS MORE LIKE SOMETHING IN THE INTRODCUTION SECTION. IN THIS SECTION, WE SHOULD NOT BE MOTIVATING THE PROBLEM BUT ANALYZING PERFORMANCE RESULTS.

On the other hand, the proposed framework achieves synchronization using the same IEEE 802.11ad waveform, shared RF hardware, and common spectral resources employed for radar sensing, thereby eliminating the need for dedicated synchronization infrastructure while maintaining low-latency operation. Within this unified ISAC framework, the bistatic UAV system operates in stripmap mode, where each IEEE 802.11ad packet has a duration of $2\,\mu s$ and is transmitted over a pulse repetition interval (PRI) of $0.66\,\mathrm{ms}$. Both UAV platforms move along the positive azimuth dimension with a total flight path length of $100\,m$, completed in $2\,s$ under a constant velocity of $50\,m/s$. Thus, a total of 3025 packets are transmitted within a CPI. Subsequently, each packet received by UAV 2 is processed to extract timing, carrier phase, and platform-state information from the data embedded in the packet. This processing takes 1.6 ms per packet and is executed in parallel upon packet reception, thereby avoiding cumulative delay across the CPI. Note that this step is not required at the UAV 1 receiver, and the received signals are sent directly to the radar signal processing block after digitization and downconversion. Further, the coarse MOCOMP applied to the raw data takes 1.1 ms for both monostatic and bistatic channels. Next, in the monostatic case, range compression requires 1.2 ms per pulse, which is processed in parallel because the pulses are independent and thus does not introduce serial latency accumulation. After all packets within the CPI are collected, azimuth compression is performed across slow-time samples along with fine MOCOMP, requiring 2.4 ms altogether. On the other hand, in the bistatic case, range compression is performed along with range offset processing, which together require 1.4 ms per pulse. The remaining processing time, which includes azimuth compression and fine MOCOMP, is similar to that of the monostatic channel.

After the 2-D SAR processing, UAV 2 performs data fusion upon receiving information from the monostatic SAR processing in the 3026th packet, and subsequently, 3D localization is performed using the proposed sphere-plane intersection algorithm. The computational time required for this final 3D localization stage is 1.1 ms. Since communication-aided synchronization and range compression are performed in parallel with packet transmission, the dominant latency component is the CPI duration. The overall end-to-end latency thus consists of the 990 ms followed by the post CPI processing stages, resulting in an approximate total latency of $994.5\ ms \simeq 0.9\ s$.

The detailed timing breakdown is summarized in Table II. The results show that packet-level pipelining ensures that communication synchronization and range compression occur concurrently once the packets are received via monostatic and bistatic channels. NOT SURE HOW THIS CONCLUSIONIS DRAWN?

TABLE II: Timing Breakdown of the Proposed ISAC-Based Bistatic Framework for a flight path duation of 2 s of both UAVs

Signal / Processing Stage	**Number of Packets**	**Duration of proposed method**	**Duration of InSAR**
IEEE 802.11ad packet transmission (PRI)	3026	0.66 ms each	0.66 ms each
Communication synchronization (UAV 2)	3026	1.6 ms each (parallel processing)	1.6 ms each (parallel processing)
Coarse MOCOMP (Monostatic/Bistatic)	3025	1.1 ms each (parallel processing)	1.1 ms each (parallel processing)
Range compression (Monostatic)	3025	1.2 ms each (parallel processing)	1.2 ms each (parallel processing)
Range compression (Bistatic)	3025	1.4 ms each (parallel processing)	1.4 ms each (parallel processing)
Azimuth compression (Monostatic/Bistatic)	1	1.2 ms	1.2 ms
Fine MOCOMP (Monostatic/Bistatic)	1	1.2 ms	1.2 ms
3D localization	1	1.1 ms	2.3 ms
Total End-to-End Latency	-	**2000.66 ms**	**2000.78 ms**

B. Two-Dimensional Localization Performance Analysis

A qualitative and quantitative evaluation of the 2-D SAR imaging performance for the proposed bistatic UAVs framework is presented in this section. Fig. 7 illustrates the focused SAR images for the three different targets considered: (i) a point scatterer on a flat ground, (ii) a mid-size automotive target placed on flat terrain, and (iii) the same automotive target positioned on an uneven terrain modeled using a DEM. For each scenario, four results are shown: ideal images without motion errors - are these ground truth iamges?, images where the simulation model is incorporated with translational and rotational motion errors, images after coarse motion compensation (MOCOMP), and images after fine MOCOMP.

Figs. 7(a)-(d) present the 2-D SAR viewgraphs of a single point scatterer for qualitative comparison. In the ideal scenario shown in Fig. 7(a), the target is accurately focused at the range and cross-range location corresponding to the ground truth values. In the presence of translational and rotational motion errors, significant defocusing is observed in Fig. 7(b). The energy spreads along the azimuth dimension due to uncompensated phase errors, resulting in peak broadening and sidelobe distortion. Fig. 7(c) shows the viewgraph after applying coarse MOCOMP where the major range migration induced by translational errors is largely compensated, restoring approximate target localization. However, residual phase errors still remain due to higher-order motion components and rotational deviations. These residual errors are effectively mitigated through fine MOCOMP based on phase gradient autofocus as shown in Fig. 7(d), resulting in a well-focused point target comparable to the ideal case. Figs. 7(e)-(h) show the results for the mid-size car on a flat terrain (zero elevation). In the ideal case (e), the distributed scattering from the extended target is clearly visible, with prominent reflections corresponding to strong scattering components such as the roof, chassis, and wheels. The spatial extent of the vehicle is accurately represented in both range and cross-range. When motion errors are introduced (f), the distributed scattering pattern becomes smeared and distorted. The defocusing is more pronounced than in the point-target case due to the interference between multiple scatterers. After coarse MOCOMP (g), the dominant geometric distortions are corrected, but minor blurring remains. Fine MOCOMP (h) restores image sharpness, demonstrating that the proposed two-stage motion compensation framework effectively preserves structural fidelity for distributed automotive targets on flat terrain.

Figs. 7(i)-(l) present the automotive target placed on an uneven terrain modeled using a DEM. Unlike the flat-terrain scenario, the target now possesses a non-zero elevation component. In the ideal processing case without motion errors (i), the target appears focused; however, its apparent range position is incorrect. This discrepancy arises because the slant range measured by the radar corresponds to the three-dimensional Euclidean distance between the UAVs and the elevated scatterer. When projected onto a two-dimensional ground plane under a flat-Earth assumption, the height component is not decoupled from the horizontal coordinates. Consequently, the target appears shifted in the range dimension relative to its true ground projection. This geometric misinterpretation persists even in the absence of motion errors. In the presence of motion errors (j), the image is further degraded due to phase distortions, compounding both geometric and focusing inaccuracies. Coarse MOCOMP (k) corrects the dominant motion-induced range shifts but cannot compensate for the inherent geometric projection error. After fine MOCOMP (l), the target is sharply focused, yet the range misplacement remains. The results in Figs. 7(i)-(l) highlight a fundamental limitation of conventional 2-D SAR imaging: while motion compensation can restore phase coherence and improve focusing quality, it cannot resolve height-induced geometric ambiguity. The 2-D SAR image inherently represents reflectivity as a function of range and cross-range under a single-look geometry. When the target lies on elevated terrain, the measured slant range does not uniquely determine its horizontal ground position unless the elevation is known or estimated. Therefore, even perfectly focused 2-D SAR images may exhibit incorrect ground-range localization for elevated targets. This limitation is addressed by the proposed bistatic tomographic reconstruction using the proposed sphere-plane intersection-based 3D localization framework, which explicitly exploits multi-view geometry to separate horizontal and vertical components of the target position.

Next, we quantitatively assess the focusing performance by computing the peak sidelobe ratio (PSLR) and integrated sidelobe ratio (ISLR) for each case in Table III.We observe that the ideal case without any motion error exhibits the best sidelobe suppression for all the targets, with PSLR values approaching the theoretical limits. The introduction of translational and rotational motion errors significantly degrade both PSLR and ISLR due to phase incoherence and energy spreading across range and azimuth dimensions. Application of coarse MOCOMP substantially improves image

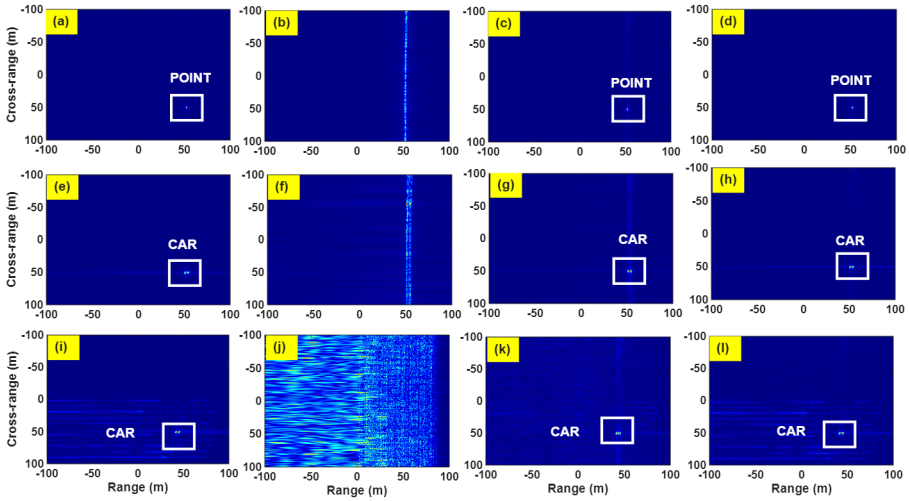


Fig. 7: SAR image formation results illustrating the impact of motion errors and motion compensation (MOCOMP). (a)–(d) Point scatterer: (a) ideal case without motion errors, (b) with translational and rotational motion errors, (c) after coarse MOCOMP, and (d) after fine MOCOMP. (e)–(h) Mid-size car located on flat terrain (zero elevation) under the same four processing conditions. (i)–(l) Automotive target placed on elevated terrain, where image formation accounts for terrain variations using a digital elevation model (DEM). The white boxes highlight the target regions. [FIGURE AXES TO BE FORMATTED]

quality by correcting dominant range migration effects; however, residual phase errors continue to limit sidelobe suppression. After fine MOCOMP using phase gradient autofocus, PSLR and ISLR values closely approach those of the ideal case, demonstrating the effectiveness of the proposed two-stage motion compensation framework. For the elevated terrain scenario, although sidelobe metrics after fine MOCOMP are comparable to the flat-terrain case, the range misregistration discussed earlier persists due to geometric projection ambiguity. This confirms that while motion compensation restores image sharpness, conventional 2-D SAR processing remains inherently incapable of resolving elevation-induced localization errors.

C. Three-Dimensional Localization Performance Analysis

To quantitatively evaluate the effectiveness of the proposed ISAC-enabled bistatic localization framework, Monte Carlo simulations are performed to analyze the 3D localization accuracy under varying values of standard deviation of platform velocity errors. The performance is assessed in terms of the RMSE of the estimated target coordinates (x, y, z) for both the proposed bistatic localization approach and the conventional InSAR-based method in Fig. 8. In addition, the analysis is carried out for both ISAC and FMCW waveforms, with and without MOCOMP, in order to investigate the combined impact of waveform and motion-error mitigation on localization performance.

Figs. 8(i), (iii), and (v) illustrate the localization RMSE corresponding to the (x, y, z) coordinates when MOCOMP is incorporated. In all the three viewgraphs, the results demonstrate that the proposed ISAC-enabled bistatic framework consistently achieves lower localization errors compared with the conventional InSAR baseline across all considered error levels. In particular, the improvement is most

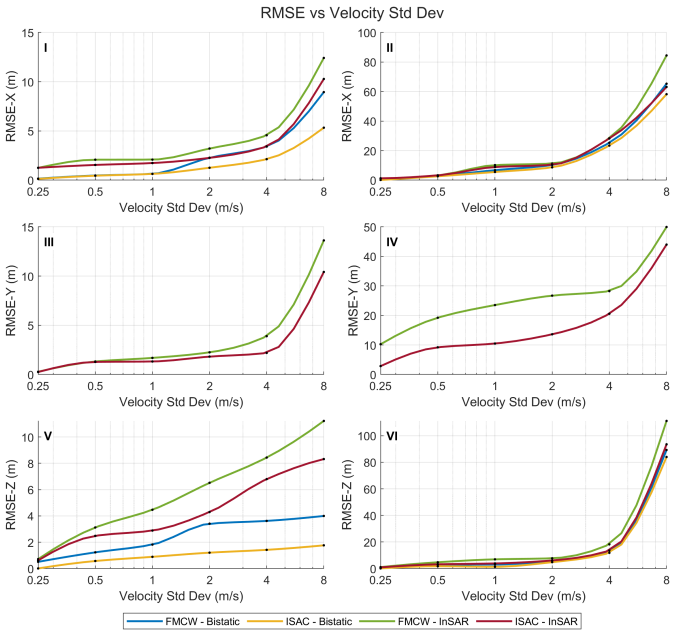


Fig. 8: RMSE comparison of estimated (x, y, z) target coordinates for the proposed bistatic localization framework and the conventional InSAR approach under varying platform velocity errors. RMSE or RMSE and sequence of x, y, z to be fixed according to the text.

significant for the estimation of the elevation (z) coordinate, where the proposed framework maintains substantially lower RMSE values for both the waveforms. This behavior arises because the proposed localization framework analytically exploits bistatic spatial diversity, thereby providing improved robustness and stability in elevation reconstruction. On the other hand, in InSAR-based localization, the elevation information is derived from interferometric phase differences,

TABLE III: Quantitative 2-D SAR Quantitative Analysis Under Different Motion Compensation Scenarios

Target Scenario	No Motion Error		With Motion Error		Coarse MOCOMP		Fine MOCOMP	
	PSLR (dB)	ISLR (dB)	PSLR (dB)	ISLR (dB)	PSLR (dB)	ISLR (dB)	PSLR (dB)	ISLR (dB)
Point on Flat Terrain	-20.75	-14.65	-11.7	7.9	-16.74	-13.92	-19.25	-14.58
Car on Flat Terrain	-20.31	-12.87	-6.79	9.70	-15.41	-11.80	-19.10	-11.24
Car on Elevated Terrain	-18.52	-10.64	-3.95	9.21	-12.71	-8.55	-17.13	-9.86

making it more sensitive to system errors. For the cross-track y coordinate, the observed RMSE values are relatively similar between the proposed framework and the InSAR approach when identical waveforms are employed. This is primarily because both methods rely on 2D SAR image formation for azimuth estimation, resulting in comparable localization behavior in the along-track dimension. In contrast, the x coordinate estimation again shows noticeable improvement for the proposed method, since the localization of the ground-range component in the InSAR approach is strongly coupled with elevation estimation errors.

Another important observation is that the ISAC waveform consistently provides lower localization errors compared with the FMCW waveform under identical operating conditions. This improvement can be attributed to the superior sidelobe suppression and higher PSLR performance of the Golay sequences within the ISAC waveform, which results in improved peak localization accuracy and reduced ambiguity in target estimation [23].

Figs. 8(ii), (iv), and (vi) present the localization performance in the absence of MOCOMP. In this scenario, all methods experience noticeable degradation in localization accuracy due to uncompensated translational and rotational motion errors in the UAV trajectories. These motion-induced deviations introduce phase distortions that adversely affect coherent SAR focusing and target localization. Consequently, the RMSE values increase significantly for all spatial coordinates compared with the motion-compensated case. Nevertheless, the proposed ISAC-based bistatic framework continues to outperform the InSAR-based approach even without motion compensation, indicating that the proposed geometric localization strategy inherently offers improved robustness against synchronization and motion uncertainties.

Overall, the results validate that the use of over-the-air ISAC-based synchronization and bistatic geometric diversity significantly enhances the robustness and accuracy of 3D target localization. The proposed framework consistently achieves lower RMSE values across all spatial coordinates, particularly under realistic motion-error conditions, thereby demonstrating its suitability for practical UAV-based bistatic SAR imaging applications.

D. Effect of Communication Modulation Schemes on Synchronization

In the second set of experiments, the impact of communication-based synchronization on localization performance is investigated across different modulation schemes. The proposed ISAC-based framework is evaluated using four modulation formats: BPSK, QPSK, 8PSK, and 16-QAM. In all of the previous sections, there has been no discussion on communication. What is the significance of the communication? For this analysis, the standard deviation of the measurement error is fixed at 0.5 unit? in order to represent a moderate error model dont understand what moderate means in this context. The RMSE values of the estimated target coordinates are computed for both the proposed 3D localization method and MOCOMP framework under each modulation scheme what are the two frameworks?. Further, we have analyzed the synchronization performance in terms of BER for different communication signal-to-noise ratio (SNR) levels that include 0 dB, 5 dB, 10 dB and 15 dB. The results show that lower-order modulation schemes provide improved localization accuracy due to their higher robustness to phase noise and synchronization errors. At a communication SNR of 15 dB, all modulation schemes exhibit very low bit-error rates (BER) of 1×10^{-3}, resulting in nearly identical average relative RMSE values of approximately 0.12 units? also what is relative rmse?. This indicates that reliable synchronization information can be extracted from the communication waveform when the BER remains sufficiently small. Further, for communication at SNR of 10 dB, lower modulation schemes such as BPSK and QPSK result in a very low BER. However, in contrast, higher-order modulation schemes such as 8PSK and 16-QAM exhibit noticeably higher BER values of 1.25×10^{-2} and 2.75×10^{-2} at the same SNR level, which consequently leads to larger localization errors. Specifically, the RMSE increases to 1.68 for 8PSK and further to 12.57 for 16-QAM, demonstrating the sensitivity of the synchronization-assisted localization process to communication reliability. missing units... the signif

The effect becomes more pronounced at the lower SNR condition of 5 dB. Under this scenario, BPSK maintains a relatively small BER of 7.50×10^{-3} and an RMSE of 0.58, indicating that the proposed localization framework remains stable even in degraded communication conditions. QPSK experiences a noticeable degradation in performance, with the BER increasing to 3.50×10^{-2} and the corresponding RMSE rising to 14.54. For higher-order modulation schemes, the BER increases significantly to 9.04×10^{-2} for 8PSK and 1.03×10^{-1} for 16-QAM. At these high error rates, the synchronization information becomes unreliable, resulting in synchronization failure **(*SF)** during the localization process. Consequently, accurate 3D target localization cannot be achieved under these conditions. At the lowest communication SNR of 0 dB, the synchronization performance degrades considerably due to the severe impact of noise on the communication link. Under this condition, BPSK exhibits a BER of 1.50×10^{-1}, resulting in a significantly larger RMSE of 4.58, which indicates notable degradation in localization accuracy. For QPSK, the BER further increases to 2.00×10^{-1},

leading to synchronization failure during the localization process. Similarly, higher-order modulation schemes such as 8PSK and 16-QAM experience very high BER values of 9.91×10^{-1} and 1.50×10^{-1}, respectively, which completely disrupt the synchronization information required for accurate localization. Consequently, synchronization failure is observed for these modulation schemes as well.

Overall, these results confirm that reliable communication-assisted synchronization plays a critical role in enabling accurate 3D localization in the proposed bistatic SAR framework. Lower-order modulation schemes such as BPSK and QPSK provide stable synchronization even under moderate SNR conditions, thereby enabling robust localization performance. In contrast, higher-order modulation schemes introduce significant synchronization uncertainty due to increased BER, which directly impacts localization accuracy. These findings demonstrate that the proposed ISAC-enabled bistatic tomographic framework effectively leverages communication signals for synchronization while maintaining reliable three-dimensional target localization, particularly when robust modulation schemes are employed.

VI. Conclusion

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TABLE IV: Performance comparison (RMSE) of different modulation schemes as a function of communication metrics

Comm SNR	BPSK				QPSK				8-PSK				16-QAM			
	BER	RMSE X	RMSE Y	RMSE Z	BER	RMSE X	RMSE Y	RMSE Z	BER	RMSE X	RMSE Y	RMSE Z	BER	RMSE X	RMSE Y	RMSE Z
15 dB	1.00×10^{-3}	0.12	0.12	0.12	1.00×10^{-3}	0.12	0.12	0.12	1.00×10^{-3}	0.12	0.12	0.12	1.00×10^{-3}	0.12	0.12	0.12
10 dB	1.00×10^{-3}	0.12	0.12	0.12	1.25×10^{-3}	0.12	0.12	0.12	1.25×10^{-2}	1.68	1.68	1.68	2.75×10^{-2}	12.57	12.57	12.57
5 dB	7.50×10^{-3}	0.58	0.58	0.58	3.50×10^{-2}	14.54	14.54	14.54	9.04×10^{-2}	*SF	*SF	*SF	1.03×10^{-1}	*SF	*SF	*SF
0 dB	1.08×10^{-1}	43.28	43.28	43.28	2.03×10^{-1}	*SF	*SF	*SF	3.34×10^{-1}	*SF	*SF	*SF	5.01×10^{-1}	*SF	*SF	*SF

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